

Practicum CFD 2026-2027

Lecturer: R. Luppés (r.luppés@rug.nl)

Practicum 2

deadline: Monday, September 21st, 17:00

- Working in pairs (groups of max. 2 students) is allowed.
- Each practicum is closed with an oral discussion supported by a (short) written report of the findings. Make the report by pen/paper (readable) or with L^AT_EX, Word, etc.
- Always use Python or Matlab/Octave for the programs that you develop.
- Save your programs in separate files, such that it is easy to demonstrate them.
- The use of AI tools such as ChatGPT is not allowed!
- Be sure that you have discussed the exercises and programs before the deadline!

Exercise 1 – Implement B3 and non-uniform grids

”Copy and Modify” the program you composed in Practicum 1 to solve the steady convection-diffusion equation

$$U \frac{d\phi}{dx} = k \frac{d^2\phi}{dx^2} \quad \text{on} \quad 0 \leq x \leq L, \quad \text{with} \quad \phi(0) = \phi_0, \quad \phi(L) = \phi_1.$$

Take into account the following:

- The 2nd order upwind method (B3) should be an additional option in case of equidistantly spaced grids.
- Very fine grids should be possible. Hence, only store the non-zero diagonals of A and use the penta-diagonal matrix algorithm PDMA as provided.
- Non-equidistantly spaced grids should be possible in combination with both methods A and B (see Section 1.5) for convection.

Exercise 2 – Theoretical analysis

- Derive for which Peclet numbers the numerical solution (equidistant grid) will be wiggle-free, when the B3 method is used together with standard (2nd order) discretisation for diffusion. Check the result using the formula on p.10. Hint: try fundamental solutions of the form r^i , and monitor the sign of r . Alternatively, study the (signs of the) coefficients.
 - Confirm your result with numerical simulations (equidistant grids). Take $U = 5$, $k = 0.01$, $L = 1$, $\phi_0 = 0$, $\phi_1 = 1$.
- For $\lambda = \frac{1}{6}$, the discretisation of $d\phi/dx$ is even 3rd-order accurate. Explain whether for ’practical’ simulations of (steady) water waves ($U > 1$, $k = 10^{-6}$) this method is really to be preferred over the B3 method, which only has 2nd order accuracy. You may use the formula on p.10 with $\lambda = \frac{1}{6}$ directly.

Exercise 3 – Accuracy study

- Show by means of a number of numerical simulations (equidistant grids) that the B3 method has indeed 2nd order accuracy. Use the analytic solution as reference solution.
- Determine by means of a number of simulations the order of accuracy of method A on non-uniform grids with a stretch rate $h_+/h_- = 0.7$ (or even $h_+/h_- < 0.5$). The purpose is to analyse the accuracy in case of ”serious” stretching, and not ”mild” stretching as e.g. in $h_+/h_- = 0.95$.

P.T.O.

- (2) Determine the eigenvalues in each case. Are they always located in the stable half plane, as they should?
- (3) Also determine the eigenvalues for method B, for a number of grids. Discuss the 'behaviour' of these eigenvalues, and their relation with the accuracy of the method.

Exercise 4 – Various discretization methods

Description

As in Practicum 1, we consider the inhomogeneous convection-diffusion equation on $0 \leq x \leq 1$, with $\phi(0) = 0$, $\phi(1) = 1$

$$\frac{d\phi}{dx} - k \frac{d^2\phi}{dx^2} = \begin{cases} 2 - 5x & , \quad 0 \leq x \leq 0.4, \\ 0 & , \quad 0.4 < x \leq 1. \end{cases}$$

Matlab file `cfid_2` solves this equation with a number of discretization methods. The solutions are shown in one combination figure – the exact solution is also indicated. Each figure shows eight smaller pictures, arranged as follows:

<i>uniform grid</i>	upwind	smart upwind	
<i>lambda schemes</i>	B3 ($\lambda = 1/2$)	QUICK ($\lambda = 1/8$)	central ($\lambda = 0$)
<i>non-uniform grid</i>	upwind	central A	central B

Two types of grid are used: uniform grids and non-uniform (stretched) grids. The 'stretched' grid possesses $N/2$ equal grid cells between 0 and $1 - 5k$, and $N/2$ equal grid cells between $1 - 5k$ and 1.

The Matlab script also gives the eigenvalues of Method B; they are displayed in Matlab's main window.

Required files: `cfid_2.m`, `cfid_upw.m`, `cfid_lam.m`, `cfid_cen.m`, `ns_exact.m`.

Questions

Compute the solutions for $k = 0.05$, $k = 0.01$ and $k = 0.001$ (i.e. $P = 1$, 5 and 50). Use a grid with $N = 20$ grid cells

- Compare the upwind results with those of the 'smart upwind' method for the same values of k . Give a personal opinion of the usefulness of the latter method in comparison with the 'standard' upwind.
- Compare the results of the lambda-schemes (see Section 1.3) with those of the upwind method; watch features like wiggle-dependency and artificial diffusion. Determine empirically for which value of k the QUICK method starts to wiggle, and explain this 'wiggle boundary' theoretically (see Exercise 3.).
- Consider the results on the stretched grid. Compare the upwind method and both generalizations (A and B) of the central method. Describe the difference between the discrete solutions. Why has the upwind solution not improved?
- On the stretched grid, determine empirically the value of k for which one of the eigenvalues of the coefficient matrix of Method B crosses the imaginary axis. Furthermore, try to find a value of k for which the matrix is approximately singular. Describe how the solution of Method B behaves at this singularity.